Ahmedabad University

ECE 501: Digital Image Processing

Week 6 Progress Report

Group 6

Vehicle Orientation Estimation Using Conventional Image Processing

Submitted to faculty: Prof. Mehul Raval

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GitHub Repository (DRASHTI-HaOBB):

https://github.com/ERYAGNIK003/DRASHTI-Dataset

Dataset Source:

DRASHTI-HaOBB – Heading-angle Oriented Bounding Box Dataset

https://zenodo.org/records/18278989

1. Work Done This Week

In Week 6, we continued investigating the high angular errors observed in the Hough Transform method during the previous experiments. The Week 5 analysis showed that Hough Transform had a higher mean angular error than PCA and contour-based estimation. Therefore, this week focused on examining the high-error cases more closely and testing a modification to the Hough line-selection procedure.

The work started with recovering the vehicle crops used in the previous experiment. Since the Week 3 results contained the predictions and error table but the complete set of vehicle crops had not been saved, we retrieved the required images directly from the remote DRASHTI-HaOBB train.zip archive. The corresponding annotations were read from train_original.zip, and the bounding-box coordinates were used to extract the vehicle regions.

All 100 vehicle crops have been saved in the Week6_Results/vehicle_crops folder. Initially a server error was returned for one image request, but the image was recovered when the request was repeated. All 100 vehicle records are contained in the final crop manifest.

Next, we selected the five vehicles with the highest angular errors from the Week 3 results. Their original crops, preprocessed edge maps, detected Hough line segments, and selected line segments were visualised. This helped us examine the line patterns that may have contributed to inaccurate orientation estimates.

Following the visual investigation, a modified Hough-based approach was used. The modification consisted of filtering short line segments, grouping the rest by their orientation and choosing the largest group of line segments in the dominant orientation group.

The parameters of the modified method were selected using a tuning subset of 70 vehicles. The remaining 30 vehicles were kept as a held-out evaluation set. The baseline and modified methods were then compared using angular error, error distribution, class-wise performance, and individual prediction changes.

The results indicated that the modified method decreased the median angular error of the held-out set, with a slight increase in the mean angular error compared with the recomputed baseline of Week 6. As a result, the modification had mixed effects and needs to be further explored.

The Week 6 workflow was:

Week 3 Results CSV → Remote Image Access → Vehicle Crop Reconstruction → High-Error Sample Visualisation → Modified Hough Implementation → Parameter Tuning → Held-Out Evaluation → Graphs and Tables → Saving Results to Google Drive

2. Methodology

2.1 Loading the Previous Results

The Week 3 CSV file was loaded into Python using Pandas. The file contained the vehicle identifiers, vehicle classes, ground-truth orientations, and previously estimated orientations from Hough Transform.

The historical angular error was recalculated using the difference between the ground-truth and predicted orientations:

$$E = |\theta_predicted - \theta_ground\ truth|$$

Both angles were represented using the project's 0°–90° orientation convention.

The saved Week 3 results were retained as a historical reference. These results were not overwritten by the new Hough predictions generated during Week 6.

2.2 Recovering the Vehicle Crops

The individual vehicle crops were reconstructed because they were not available as a complete saved set from the Week 3 experiment.

The original aerial images were accessed directly from the remote train.zip archive, while the annotation records were read from train_original.zip.

The following steps were performed:

1. Read the 100 vehicle identifiers from the Week 3 results CSV.
2. Extract the corresponding source image identifiers.
3. Locate the aerial images in the remote ZIP archive.
4. Read the corresponding vehicle annotations.
5. Extract the four oriented bounding-box corner coordinates.
6. Crop the vehicle region with a small padding of five pixels.
7. Save each crop using its original vehicle identifier.

The extracted crops were checked against the vehicle classes recorded in the Week 3 CSV. Heading information was also checked where available.



Figure 1. Examples of the vehicle crops reconstructed from the DRASHTI-HaOBB dataset.

2.3 Image Preprocessing

The same general preprocessing pipeline was used for the baseline and modified Hough experiments. The pipeline consisted of:

- Grayscale conversion
- Contrast enhancement using CLAHE
- Gaussian smoothing
- Canny edge detection
- Morphological closing

Grayscale conversion simplified the image representation. CLAHE enhanced local contrast, while Gaussian smoothing reduced small image variations. Canny edge detection was used to identify prominent boundaries, and morphological closing helped connect nearby edge fragments.

These operations produced an edge map that was supplied to the Probabilistic Hough Transform.

The preprocessing settings used in the Week 6 implementation were a 5 × 5 Gaussian kernel, Canny thresholds of 50 and 150, and a 3 × 3 morphological closing kernel.

2.4 Baseline Hough Transform

The baseline method was implemented using the Probabilistic Hough Transform available in OpenCV.

The method detected line segments in the edge map. The orientation and length of each detected segment were calculated, and a length-weighted axial mean was used to estimate the dominant direction.

The baseline parameters were:

- Hough threshold: 30
- Minimum line length: 20 pixels
- Maximum line gap: 10 pixels

The estimated axis was converted to the same 0°–90° representation used in the earlier comparison.

This baseline was important because it provided a reference for evaluating the effect of the modified line-selection procedure.

However, the recomputed Week 6 baseline was not necessarily identical to the original Week 3 implementation because the crop construction and preprocessing settings were not verified as exactly identical. The comparison between the Week 6 baseline and modified Hough method is therefore the main comparison for this week's experiment.

2.5 Modified Hough Transform

The modified approach was developed to investigate whether more selective line processing could reduce inaccurate orientation estimates.

The method involved the following steps:

1. Detect line segments using the Probabilistic Hough Transform.
2. Filter out line segments shorter than a relative length threshold.
3. Calculate the orientation of the remaining segments.
4. Group line segments according to their orientations.

5. Select the orientation group with the greatest combined line length.
6. Calculate the estimated orientation using a length-weighted axial mean.

The idea behind this modification was that short or inconsistently oriented lines might have an unnecessary influence on the estimated direction.

The selected parameters were:

Parameter	Selected Value
Hough threshold	30
Minimum line length	15 pixels
Maximum line gap	5 pixels
Minimum relative line length	0.05 of crop diagonal
Orientation bin width	20°

Table 1. Selected parameters for the modified Hough Transform.

The parameter combinations were evaluated using the tuning subset. The combination with the lowest selection score was chosen, with failed predictions penalized during parameter selection.

The selected parameters were then applied to the held-out vehicles without further tuning.

2.6 Visual Investigation of High-Error Samples

Five samples with the largest angular errors in the past were chosen.

For each vehicle, the following images were generated:

- Original vehicle crop
- Canny edge map
- All detected Hough line segments
- Line segments selected by the modified method

The purpose was to investigate whether the detected lines represented the vehicle's longitudinal direction or whether they were influenced by other edges within the crop.

The analysis also helped identify cases where the selected line group differed from the overall edge structure.

The visualizations represent the Week 6 processing and modified line selection. The angular error displayed in each title refers to the historical Week 3 result for that vehicle.

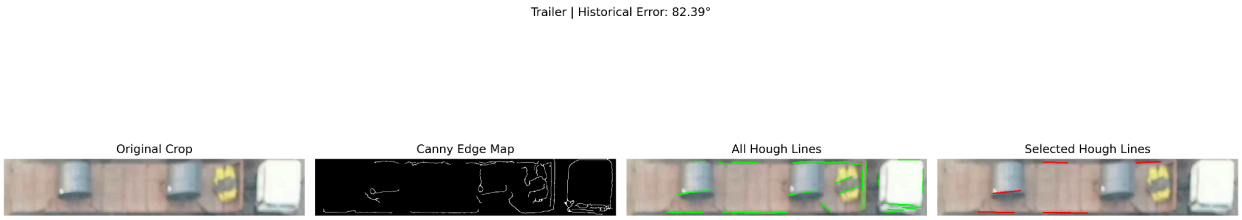


Figure 2. Hough line analysis of the Trailer sample with a historical angular error of 82.39°.

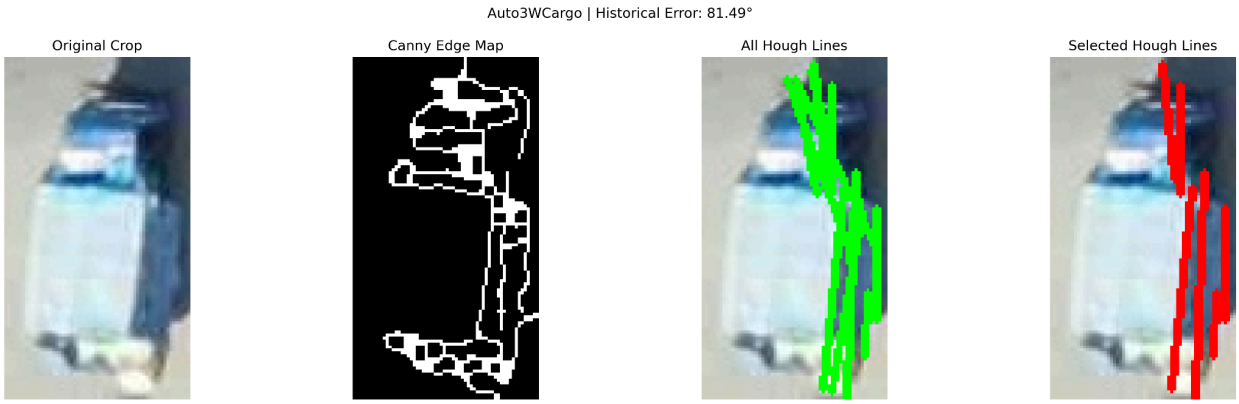


Figure 3. Hough line analysis of the Auto3WCargo sample with a historical angular error of 81.49°.

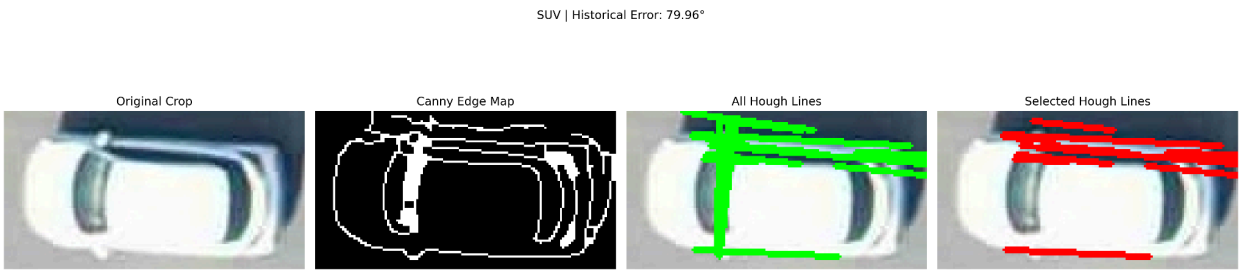


Figure 4. Hough line analysis of the SUV sample with a historical angular error of 79.96°.

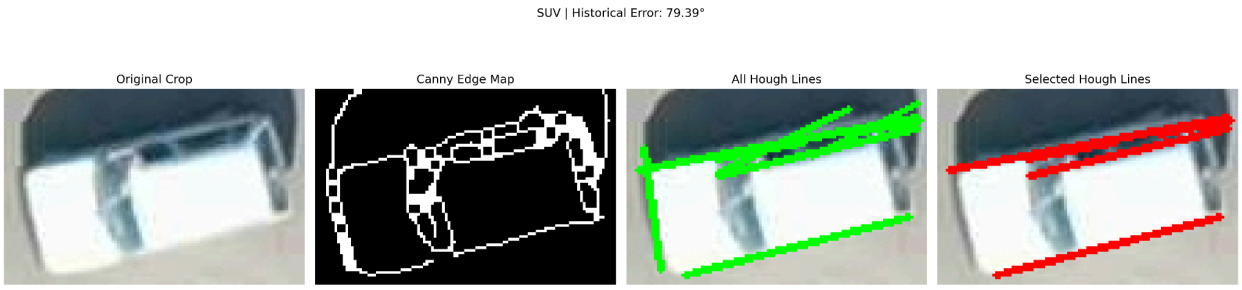


Figure 5. Hough line analysis of the SUV sample with a historical angular error of 79.39°.

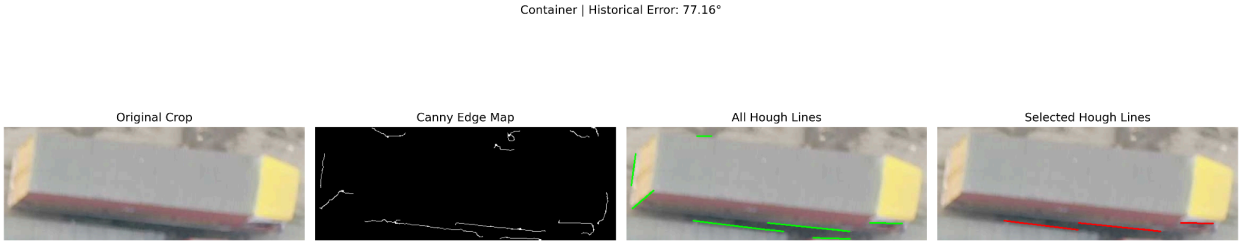


Figure 6. Hough line analysis of the Container sample with a historical angular error of 77.16°.

2.7 Experimental Evaluation

The 100 vehicle samples were divided into two subsets:

- Tuning subset: 70 vehicles
- Held-out evaluation subset: 30 vehicles

The split was made by source image so that vehicle crops from the same aerial image were kept within the same subset. A fixed random seed of 42 was used for reproducibility.

A grid of 48 parameter combinations was evaluated on the tuning subset. The best parameter combination was selected according to the defined selection score, which penalised failed predictions.

The held-out subset was then used to evaluate the original and modified Hough methods.

The following metrics were calculated:

- Number of successful predictions
- Number of failed predictions
- Mean angular error

- Median angular error
- Maximum angular error
- Standard deviation of angular error

The per-vehicle results were also compared to determine how many samples improved or worsened after the modification.

2.8 Saving the Results

The graphs, tables, crop images and summary file were saved in the Week6_Results folder of Google Drive.

The outputs saved are the vehicle crop manifest, extraction log, the experiment split, the parameter tuning table, the per-vehicle Hough results, the overall comparison table, the class-wise comparison, and the individual improvement analysis.

This organisation allows the outputs to be reviewed and reused without repeating the image retrieval and crop extraction steps.

3. Results and Observations

3.1 Recovery of the Vehicle Crops

The Week 6 workflow successfully reconstructed all 100 vehicle crops from the original dataset images and annotations.

During remote image retrieval, one request initially returned an HTTP 500 server error. The image was recovered after retrying the request, and the final crop manifest contains 100 vehicle records.

The reconstructed crops were saved under Week6_Results/vehicle_crops.

The initial extraction failure log still contains the record of the first unsuccessful request. This is an initial error record rather than evidence of a missing crop in the final manifest.

3.2 Comparison of the Original and Modified Hough Methods

The held-out evaluation set contained 30 vehicles. All three sets of predictions in the Week 6 comparison were available for these 30 samples.

Method	Successful Predictions	Failed Predictions	Mean Error	Median Error	Maximum Error	Standard Deviation
Historical Week 3 Hough	30	0	10.46°	3.83°	77.16°	16.17°
Week 6 Baseline Hough	30	0	8.01°	4.76°	82.68°	14.77°
Week 6 Modified Hough	30	0	8.81°	2.35°	84.94°	18.24°

Table 2. Overall Hough Transform comparison on the 30-vehicle held-out evaluation set.

The Week 6 modified Hough method produced a lower median angular error than the recomputed baseline, decreasing from 4.76° to 2.35°. However, the mean angular error increased from 8.01° to 8.81°, and the maximum error increased from 82.68° to 84.94°.

The overall mean error on the held-out set, therefore, did not improve with the modified method. The smaller median suggests that the typical error was smaller, but some large errors still had a significant impact on the overall performance.

The historical Week 3 Hough result on the same held-out samples had a mean error of 10.46°. However, the Week 6 baseline was recomputed using the reconstructed crops and the Week 6 processing settings. Since the original Week 3 cropping and preprocessing were not confirmed to be identical, this difference should not be interpreted as a direct improvement caused by the Week 6 modification.

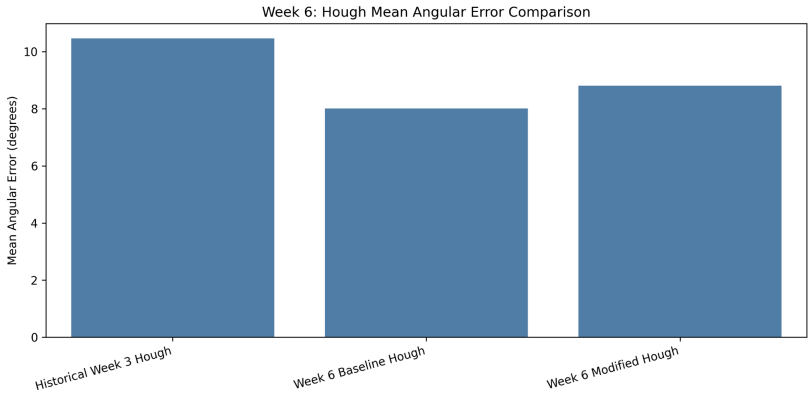


Figure 7. Comparison of mean angular error.

3.3 Error Distribution

The mean and median error results show that the modification affected the distribution of prediction errors in different ways.

The modified method had a lower median error, but its standard deviation was higher than that of the Week 6 baseline. The maximum error also increased.

This suggests that the modified line-selection procedure improved some predictions but substantially worsened others. Therefore, the result cannot be judged solely by the median.

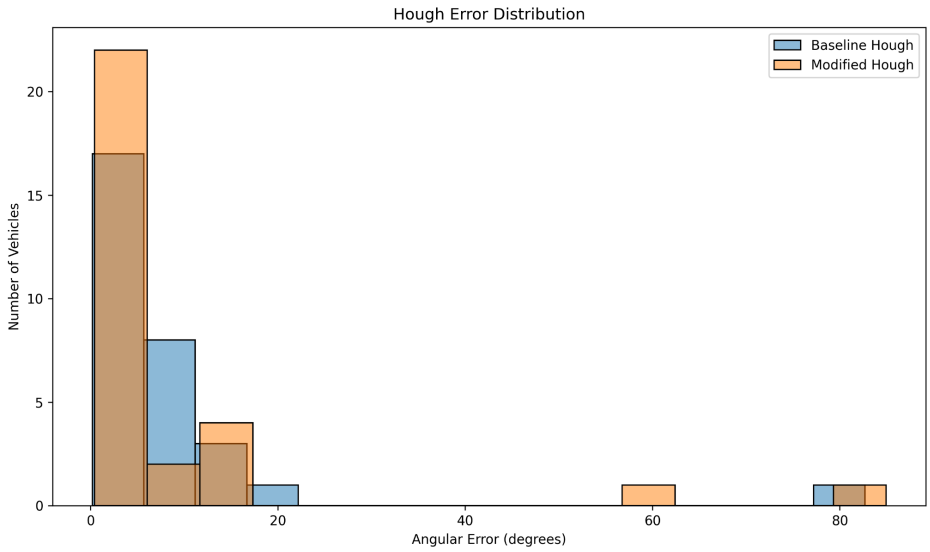


Figure 8. Distribution of angular errors for baseline and modified Hough on the held-out set.

The histogram indicates that the majority of the errors were within a relatively small range, with a small number of errors being quite large. The high-error cases are significant as they can have a significant impact on the mean angular error.

3.4 Class-Wise Performance

The held-out results were grouped by vehicle class to determine whether the modification affected different vehicle types in the same way.

Vehicle Class	Sample Count	Baseline Mean Error	Modified Mean Error	Error Reduction
Container	1	4.81°	0.49°	+4.32°
MotorCycle	12	4.69°	7.51°	-2.82°
PickUp	2	4.96°	3.02°	+1.93°
SUV	11	12.51°	9.49°	+3.02°
Sedan	2	6.72°	1.05°	+5.66°
Tanker	1	4.00°	1.29°	+2.71°
Truck	1	14.16°	59.82°	-45.66°

Table 3. Class-wise comparison of baseline and modified Hough on the held-out set.

Error reduction is calculated as baseline mean error minus modified mean error. A positive value indicates lower error with the modified method; a negative value indicates higher error.

The results show that the modification did not perform equally well across the represented vehicle classes.

The mean error decreased for Container, PickUp, SUV, Sedan, and Tanker in the held-out sample. In contrast, the mean error increased for MotorCycle and Truck.

The Truck result was particularly noticeable, with the mean error increasing from 14.16° to 59.82°. However, this class had only one vehicle in the held-out subset. The result therefore represents an individual sample rather than sufficient evidence about the performance of the method on all trucks.

Similarly, Container, PickUp, Sedan, and Tanker had very few samples. The class-wise results should be treated as preliminary observations.

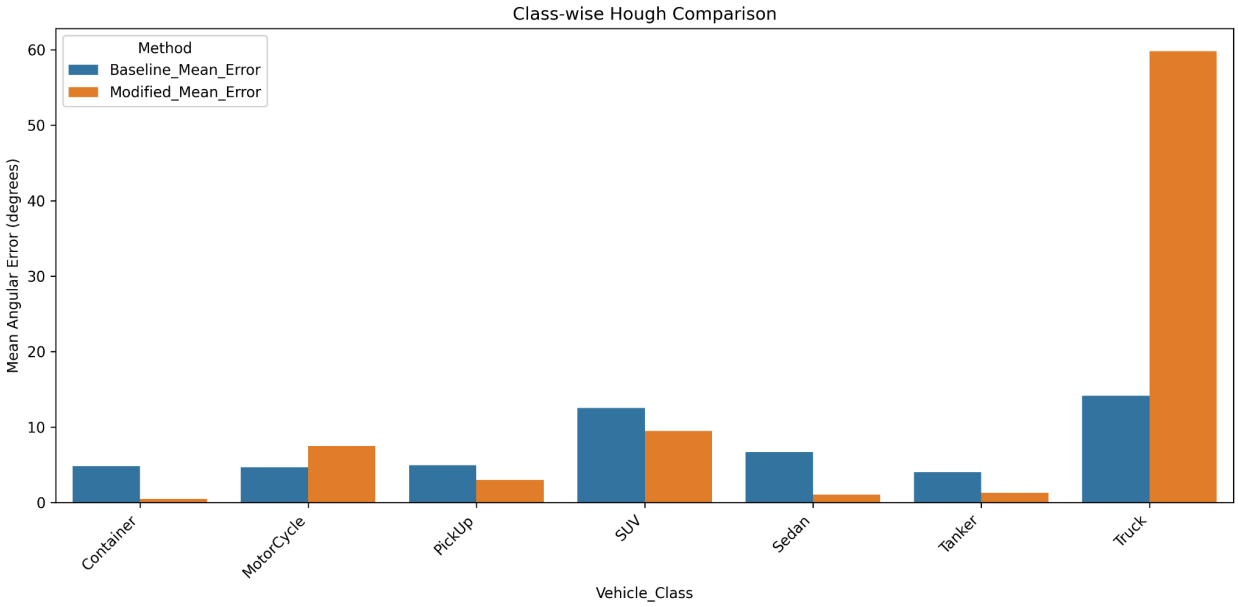


Figure 9. Class-wise mean angular error comparison between baseline and modified Hough.

3.5 Individual Prediction Changes

The individual predictions were compared to determine how many vehicles benefited from the modification.

Result	Number of Vehicles
Improved predictions	18
Worsened predictions	12
Unchanged predictions	0
Total evaluated vehicles	30

Table 4. Summary of individual prediction changes on the held-out set

The modification reduced the angular error for 18 out of 30 vehicles and increased it for the remaining 12. No prediction had exactly the same error under both methods.

Although more vehicles improved than worsened, the overall mean error still increased. This happened because some of the worsened predictions had much larger errors than the improvements achieved by other vehicles.

The average error reduction, calculated as baseline error minus modified error, was approximately -0.80°. Therefore, on average, the modified method had an error about 0.80° higher than the baseline. The median paired improvement was approximately 0.78°, showing that the typical change was more favourable than the mean change suggested.

These results indicate that the modification may help some samples but is not yet reliable enough to be considered a clear overall improvement.

3.6 Parameter Tuning Results

The modified Hough method was tuned using 70 vehicles. The selected parameters were:

- Hough threshold: 30
- Minimum line length: 15 pixels
- Maximum line gap: 5 pixels
- Minimum relative line length: 0.05
- Orientation bin width: 20°

The modified method achieved a mean error of 3.87° on the tuning subset. However, its mean error increased to 8.81° on the held-out subset.

The tuning error is not an independent measure of final performance because the parameters were selected using those samples. The difference between tuning and held-out results suggests that the selected parameter combination did not generalise equally well to unseen samples.

The tuning results were therefore used only to select parameters, while the held-out results were used for the main performance comparison.

3.7 Investigation of Possible Hough Error Sources

The visualisations of high-error vehicle samples show the original crops, their Canny edge maps, all detected Hough line segments, and the line segments selected by the modified method.

These visualisations show that the edge maps can contain several line structures at different orientations. The detected lines do not always form a single clearly dominant direction.

Possible reasons for the high errors include:

1. Multiple prominent edges: Windows, roofs, and other vehicle structures can produce line segments at different angles.
2. Background interference: Other structures within the crop may introduce lines that do not belong to the vehicle's main axis.
3. Vehicle geometry: Some vehicle shapes may not have a clear set of long edges aligned with the longitudinal direction.
4. Weak or fragmented edges: Low resolution, blur, and incomplete boundaries can affect the detected line segments.
5. Line selection: The dominant group of detected segments may not correspond to the vehicle's actual longitudinal axis.

These are possible explanations based on the behaviour of the method. The visualisations help investigate the line-selection problem, but they do not confirm that one specific factor caused every high-error prediction.

4. Challenges Faced

One challenge in Week 6 was getting the aerial images straight from the remote ZIP file. An initial image request returned an HTTP 500 server error. The request was repeated and the final crop manifest included all 100 vehicle crops.

Another challenge was ensuring that the reconstructed crops corresponded to the correct annotations. The vehicle identifiers and class information were checked to reduce the risk of matching an image with an incorrect annotation.

The main technical challenge was selecting Hough line segments that represent the vehicle's longitudinal axis. A vehicle image may contain many lines, and the longest or most prominent group does not always correspond to the correct orientation.

The parameter tuning results also showed that performance on the tuning subset was not directly reflected in the held-out results. The modified method performed well on the tuning data, but it did not reduce the mean angular error on the held-out set.

Finally, the class-wise analysis was limited by the small number of samples in certain categories. The high error observed for Truck, for example, was based on only one held-out vehicle and cannot be generalised to all trucks.

5. Current Status

The Week 6 work extended the previous error analysis by investigating Hough line detection and testing a modified line-selection procedure.

The work completed includes:

- Recovering and saving 100 vehicle crops.
- Identifying five high-error vehicle samples.
- Generating visualisations of the crops, edge maps, and detected Hough lines.
- Implementing a modified Hough line-selection method.
- Testing 48 parameter combinations on the tuning subset.
- Comparing the baseline and modified methods on 30 held-out samples.
- Generating overall and class-wise comparison tables.
- Analysing individual predictions that improved or worsened.
- Saving the results in the Week6_Results folder.

The held-out results showed that the modified Hough method reduced the median error from 4.76° to 2.35°, but increased the mean error from 8.01° to 8.81°. Therefore, the modification produced mixed results rather than a clear overall improvement.

The results provide a better understanding of the limitations of the Hough Transform and establish a basis for further improvements in the remaining weeks.

6. Work to Be Done Next Week

In the next stage, we plan to focus on the final goal of the project: estimating the direction in which the front of a vehicle is pointing.

The planned work includes:

- Reviewing the axis estimates produced by PCA, Hough, and contour-based methods.
- Preserving the full 0°–180° undirected vehicle axis instead of relying only on the folded 0°–90° representation.

- Aligning vehicle crops using the estimated axis.
- Investigating visible and geometric features that may help distinguish the front from the rear.
- Developing an initial conventional front/back decision method.
- Comparing front/back predictions with the original dataset heading annotations.
- Documenting cases where the front and rear cannot be distinguished reliably.

This will extend the axis-estimation work completed during the first six weeks toward the final objective of estimating the full vehicle heading in the 0°–359° range.

7. References

[1] Y. Bhavsar, M. Zaveri, M. Raval, K. R. Patel, and S. B. Zaveri, "Descriptor: Drone Nadir-view Annotated Images of Vehicles Detection Dataset for India with Heading-angle Oriented Bounding Box (DRASHTI-HaOBB)," *IEEE Data Descriptions*, 2026. Available: https://ieeexplore.ieee.org/document/11421776.

[2] Y. Bhavsar, M. Zaveri, M. Raval, S. Zaveri, and Ahmedabad University, "DRASHTI-HaOBB: Drone nadir-view Annotated Images of Vehicles Dataset for India - Heading-angle Oriented Bounding Box," *Zenodo*, 2026. Available: https://zenodo.org/records/18278989.